

A Framework for Detecting Micro-communities on Popular Subreddits

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1 Introduction

Online platforms like Reddit have become important spaces for people to express their opinions and engage in conversations on a wide range of topics, from politics and pop culture to science and social issues. Within these platforms, subreddits serve as focused communities where users share content and discuss specific subjects. But even within a single subreddit, people don't always agree. There are often smaller groups of users who hold different opinions or bring unique perspectives to the same issue, what we refer to as micro-communities.

While a lot of research on public opinion or community behavior relies on structured data like surveys or network connections, text-based, unstructured data like Reddit posts and comments is still not fully explored for this purpose. This kind of data may seem messy, but it can hold valuable insights into how people think, feel, and interact.

In this project, we focus on understanding micro-communities by analyzing large-scale unstructured text from Reddit. Using the r/climatechange subreddit as our main example, we apply natural language processing and clustering techniques to identify groups of users with similar language, tone, or opinions. We also look at how these groups shift over time, aiming to uncover patterns in opinion dynamics, polarization, and narrative changes that wouldn't be obvious just from reading individual posts.

2 Methods

The general framework is composed of 4 key steps and shown in Figure 1. Within each step, practitioners make domain-specific choices that optimizes for their research objective. We will discuss each step in detail in the following sections.

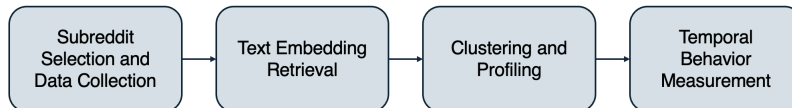


Fig. 1: **Process Framework** Some other caption stuff here.

2.1 Subreddit Selection and Data Collection

The first step involves selecting a community of interest. While this framework is directly applicable to Reddit, it can also be adapted for other forum-based platforms. For Reddit, a subreddit serves as the natural boundary for initial community definition, representing users who share a common interest around a particular topic.

Practitioners select a subreddit based on its relevance to the research question and the expected diversity of opinions within the community. Once selected, posts and comments are collected over a specified time window using Reddit’s public API or other data collection tools. Metadata such as author IDs, timestamps, upvotes, and thread structures are preserved to facilitate later analyses.

2.2 Text Embedding Retrieval

After data collection, the next step is to transform raw text into a format suitable for quantitative analysis. We apply state-of-the-art text embedding models that encode semantic, stylistic, and tonal information from posts and comments into dense vector representations.

Practitioners can choose from a variety of embedding methods depending on the task (e.g., Sentence-BERT, OpenAI embeddings, or domain-adapted transformers). Preprocessing steps typically include text cleaning, lowercasing, and removal of stopwords or noise, but care must be taken not to eliminate meaningful content.

The result is a high-dimensional dataset where each user’s language can be quantitatively compared to others. For our demonstration of the framework, we used OpenAI’s text embedding large model[1].

2.3 Clustering and Profiling

With embeddings prepared, the third step involves clustering users into micro-communities. The goal is to group users who express similar ideas, emotions, or narratives.

Clustering methods such as K-means, DBSCAN, or spectral clustering can be employed based on the structure of the embedding space and the desired granularity. Profiling each cluster involves interpreting common linguistic patterns, shared themes, or sentiment profiles among its members. This step provides a foundational map of the different micro-communities within the broader subreddit.

There are several things to consider in this phase. First, determining the number of clusters will be dependent on the domain and the context in which the process is being used. Second, the desired granularity will determine the temporal nature of the data. For example, when practitioners do not wish to aggregate embeddings, they may choose to use the most granular measure such as day or week. Aggregation on longer periods, like months or years, are best suited for interpretability and generally easier to visualize. In instances where aggregation is needed, we recommend using standard mean or median aggregation.

2.4 Temporal Behavior Measurement

Finally, we examine how micro-communities evolve over time. By segmenting the dataset into temporal windows (e.g., monthly, quarterly), we track changes in community membership, language patterns, and thematic emphasis.

Temporal measurement allows researchers to study dynamics such as polarization, convergence/divergence of opinions, narrative shifts, and the impact of external events (e.g., news, policy changes) on community behavior.

This step transforms a static snapshot of micro-communities into a dynamic understanding of how public discourse evolves within a shared online space.

3 Results

In this section, we present the results of applying our framework to one year of Reddit data from the r/climatechange subreddit. To structure the analysis, we aggregated user text embeddings at the monthly level. As a result of this aggregation, we restricted our sample to users who were active for at least six months within the one-year period, ensuring sufficient longitudinal representation.

A few additional methodological choices are important to highlight. First, the network formation criteria is defined as user connections based on co-appearance within the same post. That is, users who commented or posted in the same thread during a given month were considered to have a connection for that month’s network. Second, for interpretability, we focused on identifying three clusters per month. In visualizations, the largest cluster is consistently colored red, and the smallest cluster is colored purple.

Using this process, we generated 12 monthly networks, each colored by cluster membership.

Figure 2 shows a Sankey diagram visualizing the movement of individuals across clusters over time, as well as users who drop out of clustering (i.e., users who become inactive or no longer meet the co-appearance criteria).

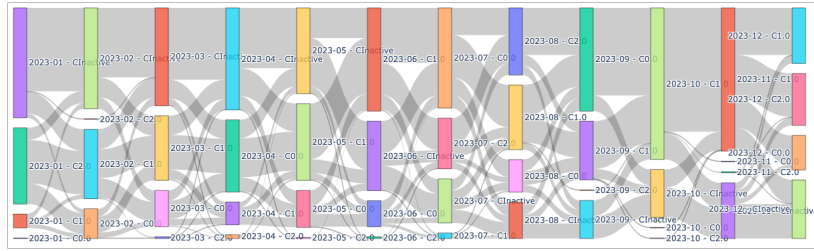


Fig. 2: Sankey Plot Some other caption stuff here.

Figure 3 shows the monthly networks of users in r/climatechange over the course of 2023, with nodes colored according to their cluster membership. Each node represents a user, and edges represent co-appearances within the same post

for a given month. The biggest cluster for every month is colored in red, and the smallest cluster is colored in purple.

Several interesting patterns emerge from this analysis. Across most months, a single large cluster dominates the network suggesting the presence of a core group of users who exhibit relatively homogeneous language patterns or thematic focus for the given month. Second, network density substantially varies over months, suggesting that individuals either engage in long periods of increased activity or intermittently engage in topics of relevance to them.

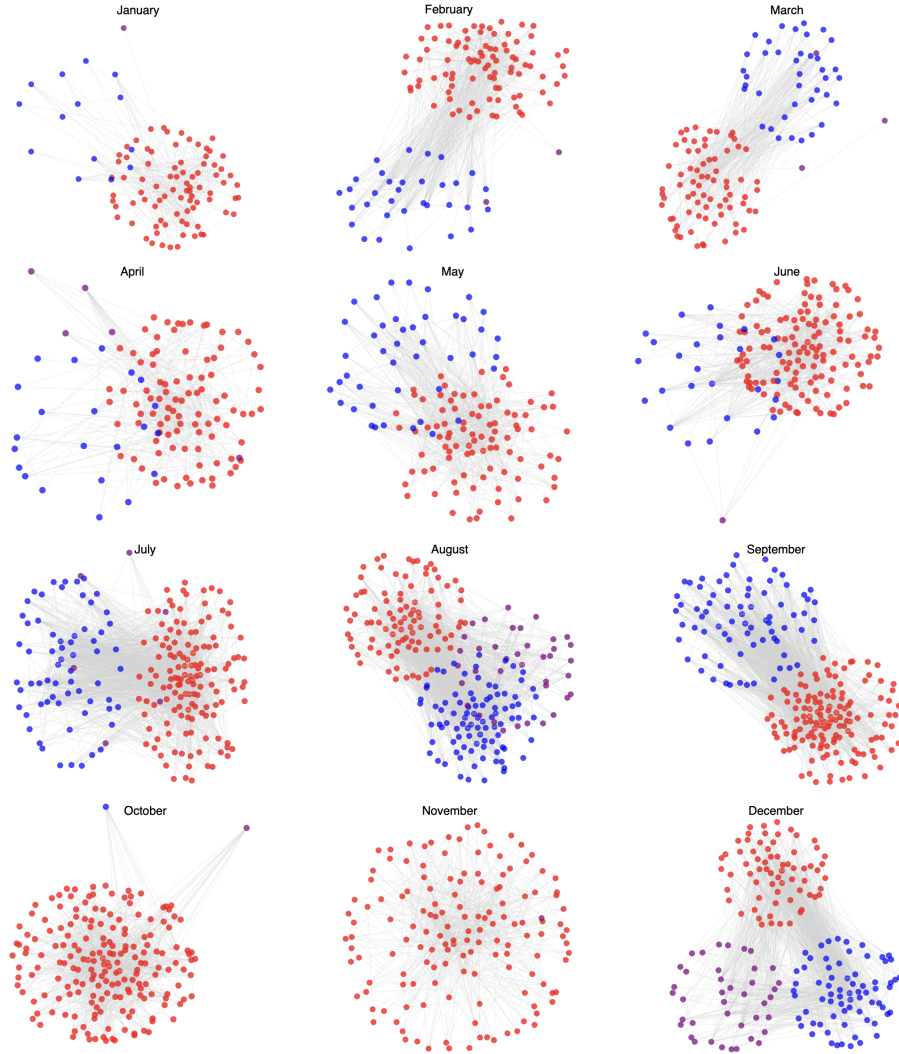


Fig. 3: **Network Graphs** Some other caption stuff here.

To characterize the topical focus of each cluster, we performed keyword extraction using both TF-IDF and KeyBERT. While TF-IDF results were some-

times noisy, KeyBERT provided more interpretable insights. Across different months, we found that clusters captured distinct thematic areas. Some clusters reflected ecological concerns, such as beekeeping and environmental conditions (e.g., "worried bees," "sugar syrup"). In contrast, others captured discussions around climate science and mitigation strategies (e.g., "climate models," "carbon need," "future electricity demand"). Additionally, certain smaller clusters focused on niche events like environmental film festivals or administrative topics related to post moderation. Overall, this profiling suggests that while a core group of users consistently engaged in substantive climate discourse, peripheral groups periodically formed around specific topics or events.

4 Conclusion

In this project, we explored how user conversations on Reddit can be used to understand the formation and evolution of micro-communities over time. Using the r/climatechange subreddit as a case study, we applied text embeddings, clustering, and network analysis to track how users group based on the language they use and the topics they focus on. Our results showed that while a core group of users stays consistently active around climate discussions, smaller groups can form around specific events or other side conversations. Overall, this approach shows that even messy, unstructured text data can be a powerful tool for studying how online communities grow, shift, and organize themselves over time.

References

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